CERTIFIED FINITE POSITIVITY BANDS FOR A RIEMANN Ξ-DEFECT KERNEL VIA ENDPOINT MOMENT BOUNDS

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ABSTRACT. Let $\Xi(z) = \xi(\frac{1}{2} + iz)$ and let Φ denote the positive even Riemann theta kernel in its full-line Fourier representation. We study the two-variable cosine transform

$$C_y(x) = \iint_{\mathbb{R}^2} (u+v)^2 \operatorname{sinhc}(y(u+v))\Phi(u)\Phi(v) \times \cos(x(u-v))\, \mathrm{d}u\, \mathrm{d}v, \qquad 0 \le y \le \tfrac{1}{2}.$$

The Hermite–Biehler defect of $E = \Xi + i\Xi'$ equals $2yC_y(x)$. We derive exact endpoint formulas for the even moments of the positive measure defining C_y, bound the normalized moments by strict cross-endpoint ratios, and integrate a global degree-ten Taylor minorant for cosine. Outward-rounded Arb computation then proves

$$C_y(x) > 0 \qquad \left(0 \le y \le \tfrac{1}{2},\ |x| \le R_{10}\right),$$

where

$$7.0362433 < R_{10} < 7.0362434.$$

The certificate uses no zero enumeration. The result is finite: it neither proves the Riemann Hypothesis nor improves existing finite-height zero verification. We identify the unproved all-orders estimate that the cross-endpoint method would require for unbounded positivity bands.

1. Introduction

Write

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \Xi(z) = \xi\left(\frac{1}{2} + iz\right). \tag{1}$$

The completed function satisfies

$$\xi(s) = \xi(1-s), \tag{2}$$

and Ξ is real entire and even. The Jacobi transformation gives the full-line representation

$$\Xi(z) = \int_{\mathbb{R}} \Phi(u)e^{izu}\, \mathrm{d}u, \tag{3}$$

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where, in the normalization used here,

$$\Phi(u) = \sum_{n=1}^{\infty} \left(4\pi^2 n^4 e^{9|u|/2} - 6\pi n^2 e^{5|u|/2}\right) e^{-\pi n^2 e^{2|u|}}. \tag{4}$$

For $u \geq 0$, each summand is positive because

$$4\pi^2 n^4 e^{9u/2} - 6\pi n^2 e^{5u/2} = 2\pi n^2 e^{5u/2}\left(2\pi n^2 e^{2u} - 3\right) > 0.$$

The Jacobi transformation makes the extension in (4) even and smooth. The kernel also satisfies a bound

$$0 < \Phi(u) \leq C e^{9|u|/2} \exp\left(-\frac{\pi}{2} e^{2|u|}\right) \tag{5}$$

for a fixed $C > 0$.

Moment inequalities, Fourier criteria, and Hermite–Biehler formulations for Ξ have a substantial history. Csordas, Norfolk, and Varga proved Pólya–Turán inequalities for Taylor coefficients [3]. Csordas and Varga proved strict concavity of $t \mapsto \log \Phi(\sqrt{t})$ and obtained broad moment inequalities [4]. Their later work developed Hermite–Biehler and generalized Laguerre criteria [5, 6]. Coffey and Csordas gave another proof of logarithmic concavity and related moment inequalities [2], while Csordas studied positive-definite associated kernels and isolated global positivity conditions equivalent to the Riemann Hypothesis [1].

The contribution here is narrower. We combine moments at $y = 0$ and $y = \frac{1}{2}$ to obtain one explicit polynomial lower bound for a two-variable defect kernel, uniformly over the closed y-interval. The endpoint moments are values of finite combinations of derivatives of ξ at 0 and $\frac{1}{2}$, so Arb can certify them without locating zeros. The first positive root of the resulting polynomial gives the stated finite band.

The zero consequence is not a record. Platt and Trudgian rigorously verified the Riemann Hypothesis through height $3 \cdot 10^{12}$ [7]. The interest of the present result lies in the moment certificate and its uniform kernel inequality, rather than the small numerical height of the corollary.

2. The defect kernel

Set

$$\operatorname{sinhc} t = \begin{cases} \sinh(t)/t, & t \neq 0, \\ 1, & t = 0. \end{cases}$$

For $p = u + v$ and $q = u - v$, define

$$W_y(u,v) = p^2 \operatorname{sinhc}(yp)\Phi(u)\Phi(v), \qquad 0 \leq y \leq \frac{1}{2}, \tag{6}$$

and

$$C_y(x) = \iint_{\mathbb{R}^2} W_y(u,v)\cos(xq)\,\mathrm{d}u\,\mathrm{d}v. \tag{7}$$

For $y > 0$, this equals the form

$$C_y(x) = \iint_{\mathbb{R}^2} \frac{p \sinh(yp)}{y} \Phi(u)\Phi(v) \cos(xq) \,\mathrm{d}u \,\mathrm{d}v.$$

Estimate (5) shows that

$$|p|^a |q|^b e^{|p|/2} \Phi(u)\Phi(v)$$

is integrable for every fixed pair of nonnegative integers a, b. Thus all integrals and derivatives used below converge absolutely and locally uniformly. Fubini's theorem, differentiation under the integral, and the limits at $y = 0$ and $y = \frac{1}{2}$ are valid.

Lemma 2.1 (Defect identity)**.** *Let*

$$E(z) = \Xi(z) + i\Xi'(z), \qquad E^{\#}(z) = \overline{E(\bar{z})}.$$

Then

$$|E(x + iy)|^2 - |E^{\#}(x + iy)|^2 = 2yC_y(x) \tag{8}$$

for real x and $0 \le y \le \frac{1}{2}$.

Proof. Put $s = \frac{1}{2} - y$ and

$$P(s, x) = \xi(s + ix)\xi(s - ix) = |\Xi(x + iy)|^2.$$

Multiplying the Fourier integrals and symmetrizing under $u \leftrightarrow v$ and $(u, v) \mapsto (-u, -v)$ gives

$$P(s, x) = \iint_{\mathbb{R}^2} \Phi(u)\Phi(v) \cosh(yp) \cos(xq) \,\mathrm{d}u \,\mathrm{d}v. \tag{9}$$

Consequently,

$$\frac{\partial P}{\partial y}(s, x) = \iint p \sinh(yp)\Phi(u)\Phi(v) \cos(xq) \,\mathrm{d}u \,\mathrm{d}v = yC_y(x). \tag{10}$$

Because Ξ is real entire, $E^{\#} = \Xi - i\Xi'$. Direct expansion yields

$$|E|^2 - |E^{\#}|^2 = -4 \operatorname{Im}(\Xi'\overline{\Xi}) = 2 \frac{\partial}{\partial y} |\Xi(x + iy)|^2.$$

Equation (10) proves (8). □

At $y = 0$, differentiation of (10) gives

$$C_0(x) = 2 \left(\Xi'(x)^2 - \Xi(x)\Xi''(x) \right). \tag{11}$$

3. Endpoint moments

For $j \geq 0$, define the raw moments

$$A_{2j}(y) = \iint_{\mathbb{R}^2} W_y(u,v) q^{2j} \,\mathrm{d}u\,\mathrm{d}v. \tag{12}$$

Then

$$C_y^{(2j)}(0) = (-1)^j A_{2j}(y), \tag{13}$$

and $A_{2j}(y) > 0$.

Define

$$H_{2j}(s) = \sum_{\ell=0}^{2j} (-1)^\ell \binom{2j}{\ell} \xi^{(\ell)}(s) \xi^{(2j-\ell)}(s). \tag{14}$$

Lemma 3.1 (Moment identities)**.** *For $y > 0$,*

$$A_{2j}(y) = -\frac{H'_{2j}(\frac{1}{2} - y)}{y}. \tag{15}$$

The continuous endpoint values are

$$A_{2j}(0) = H''_{2j}\left(\frac{1}{2}\right), \qquad A_{2j}\left(\frac{1}{2}\right) = -2H'_{2j}(0). \tag{16}$$

Proof. Analytic continuation of (3) gives

$$\xi(s) = \int_{\mathbb{R}} \Phi(u) e^{(s-1/2)u} \,\mathrm{d}u.$$

Expanding $(u-v)^{2j}$ proves

$$H_{2j}(s) = \iint_{\mathbb{R}^2} e^{(s-1/2)p} q^{2j} \Phi(u)\Phi(v) \,\mathrm{d}u\,\mathrm{d}v.$$

Equivalently, Leibniz differentiation gives

$$\partial_x^{2j} P(s,x)\Big|_{x=0} = (-1)^j H_{2j}(s).$$

Since $C_y = -P_s/y$, equations (13) and (15) follow. The functional equation implies

$$H_{2j}(1-s) = H_{2j}(s), \qquad H'_{2j}(1/2) = 0.$$

Taylor expansion at $1/2$ gives the first endpoint formula; substituting $y = 1/2$ gives the second. □

Lemma 3.2 (Strict raw-moment monotonicity)**.** *For every $j \geq 0$,*

$$0 \leq y_1 < y_2 \leq \frac{1}{2} \implies A_{2j}(y_1) < A_{2j}(y_2).$$

Consequently, with $m_{2j}(y) = A_{2j}(y)/A_0(y)$,

$$\frac{A_{2j}(0)}{A_0(1/2)} < m_{2j}(y) < \frac{A_{2j}(1/2)}{A_0(0)} \tag{17}$$

for every $y \in [0, 1/2]$ and $j \geq 1$.

Proof. For $p \neq 0$,

$$\operatorname{sinhc}(yp) = \sum_{r=0}^{\infty} \frac{y^{2r} p^{2r}}{(2r+1)!}$$

is strictly increasing for $y \geq 0$. The remaining factor in (12) is nonnegative and is positive away from measure-zero lines. This proves strict monotonicity. Applying the endpoint bounds separately to the numerator and denominator gives (17). Equality would require the numerator and denominator to attain opposite endpoints at the same value of y, so both inequalities are strict even when y is an endpoint. □

4. The degree-ten lower polynomial

Lemma 4.1 (Global cosine minorant)**.** *For every real $r \neq 0$,*

$$\cos r > T_{10}(r) := 1 - \frac{r^2}{2!} + \frac{r^4}{4!} - \frac{r^6}{6!} + \frac{r^8}{8!} - \frac{r^{10}}{10!}. \tag{18}$$

Proof. For $g = \cos - T_{10}$,

$$g^{(10)}(r) = 1 - \cos r \geq 0,$$

and $g^{(k)}(0) = 0$ for $0 \leq k \leq 9$. Repeated integration proves $g(r) \geq 0$ for $r \geq 0$. The inequality is strict for $r > 0$ because $1 - \cos r$ does not vanish identically on any interval $(0, r)$. Evenness handles negative r. □

For the five moments used below, set

$$\begin{aligned} B_2 &= \frac{A_2(1/2)}{A_0(0)}, \quad B_4 = \frac{A_4(0)}{A_0(1/2)}, \\ B_6 &= \frac{A_6(1/2)}{A_0(0)}, \quad B_8 = \frac{A_8(0)}{A_0(1/2)}, \\ B_{10} &= \frac{A_{10}(1/2)}{A_0(0)}. \end{aligned} \tag{19}$$

Define

$$P_\theta(x) = 1 - \frac{B_2 x^2}{2!} + \frac{B_4 x^4}{4!} - \frac{B_6 x^6}{6!} + \frac{B_8 x^8}{8!} - \frac{B_{10} x^{10}}{10!}. \tag{20}$$

Proposition 4.2 (Uniform polynomial lower bound)**.** *For $0 \leq y \leq \frac{1}{2}$ and $x \neq 0$,*

$$\frac{C_y(x)}{A_0(y)} > P_\theta(x). \tag{21}$$

Proof. Normalize $W_y \, \mathrm{d}u \, \mathrm{d}v$ by $A_0(y)$ and apply Lemma 4.1. Strictness holds because $xq \neq 0$ on a set of positive measure. We obtain

$$\frac{C_y(x)}{A_0(y)} > 1 - \frac{m_2(y)x^2}{2!} + \frac{m_4(y)x^4}{4!} - \frac{m_6(y)x^6}{6!} + \frac{m_8(y)x^8}{8!} - \frac{m_{10}(y)x^{10}}{10!}.$$

For negative terms, use the upper bounds in (17); for positive terms, use the lower bounds. This gives (21). □

5. Certified endpoint arithmetic

The certificate evaluates (16) from Taylor balls for

$$\xi(s) = (s-1)\pi^{-s/2}\Gamma(1+s/2)\zeta(s), \tag{22}$$

which is algebraically identical to (1) and analytic at both centers 0 and 1/2.

Table 1 lists ball centers. At 269-bit Arb precision, the largest radius among the twelve displayed values is $1.37 \cdot 10^{-67}$.

Table 1. Certified endpoint-moment ball centers.

Moment	$A_{2j}(0)$	$A_{2j}(1/2)$
A_0	0.02283966166888743667469247	0.02309570896612103381431025
A_2	0.001890366586781360982949785	0.001909616896683607096775872
A_4	0.000459720156209605774158694	0.000464013763611390134012507
A_6	0.000182390894103604245533460	0.000183966337058884295480848
A_8	0.000099156363571899705593806	0.000099954209940825929544209
A_{10}	0.000067852553023503413540728	0.000068364291172414501807343

The resulting coefficient balls have centers

$$\begin{aligned} B_2 &= 0.0836096840823574320891615416\ldots, \\ B_4 &= 0.0199050029979147510756695686\ldots, \\ B_6 &= 0.00805468748731450212309766654\ldots, \\ B_8 &= 0.00429328078723851344898443484\ldots, \\ B_{10} &= 0.00299322696472082444610981628\ldots. \end{aligned} \tag{23}$$

Put $Q(t) = P_\theta(\sqrt{t})$. On $0 \le t \le 51$, the degree-four Bernstein coefficient balls for Q' have centers

$$\begin{aligned} &-0.0418048420411787160, \quad -0.0206557763558942930, \\ &-0.0140554899445716895, \quad -0.00787928068151705499, \\ &-0.0159039318562651082. \end{aligned} \tag{24}$$

Every complete ball is negative. Since Bernstein basis functions are nonnegative on the unit interval, $Q'(t) < 0$ for $0 \le t \le 51$.

For $t = 51 + r$, write

$$Q'(51+r) = \sum_{j=0}^{4} c_j r^j.$$

The coefficient-ball centers are

$$\begin{aligned} c_0 &= -0.0159039318562651082, \\ c_1 &= -0.000629384405862592411, \\ c_2 &= -0.0000327586169268804792, \\ c_3 &= -0.000000415429458942178156, \\ c_4 &= -0.000000004124265548832706. \end{aligned} \tag{25}$$

Again every complete ball is negative, so $Q'(t) < 0$ for $t \geq 51$. Hence

$$Q'(t) < 0 \qquad (t \geq 0). \tag{26}$$

The exact rational evaluation points satisfy the certified inequalities

$$P_\theta(7.0362433) > 3.7223537277 \cdot 10^{-9} > 0, \tag{27}$$

$$P_\theta(7.0362434) < -1.7438217651 \cdot 10^{-8} < 0. \tag{28}$$

Because $Q(0) = 1$, (26) holds, and the leading coefficient of Q is negative, P_θ has exactly one positive zero. Denote it by R_{10}. Then

$$7.0362433 < R_{10} < 7.0362434, \qquad R_{10} = 7.03624331759098920168\ldots. \tag{29}$$

Theorem 5.1 (Certified finite positivity band)**.** *For every* $0 \leq y \leq \frac{1}{2}$*,*

$$C_y(x) > 0 \qquad (|x| \leq R_{10}). \tag{30}$$

Proof. For $0 < |x| < R_{10}$, Proposition 4.2 and $P_\theta(x) > 0$ prove the claim. At $|x| = R_{10}$, strictness in (21) gives

$$\frac{C_y(R_{10})}{A_0(y)} > P_\theta(R_{10}) = 0.$$

At $x = 0$, $C_y(0) = A_0(y) > 0$. Evenness in x completes the proof. □

6. Finite zero consequence

Corollary 6.1. *If* ρ *is a nontrivial zero of* ζ *with* $|\operatorname{Im}\rho| \leq R_{10}$*, then*

$$\operatorname{Re}\rho = \frac{1}{2}.$$

Proof. At a zero of Ξ, the two functions E and $E^\#$ have the same modulus, so the left side of (8) vanishes. If $\rho = \beta + i\gamma$ with $\beta < 1/2$, then

$$z = \gamma + i(1/2 - \beta)$$

satisfies $\Xi(z) = \xi(\rho) = 0$ and $0 < \operatorname{Im} z < 1/2$. For $|\gamma| \leq R_{10}$, Theorem 5.1 and (8) make the defect positive, a contradiction. If $\beta > 1/2$, the functional equation and conjugation give the zero $1 - \bar{\rho}$, with the same ordinate and real part below $1/2$. □

At a real zero x of Ξ, equation (11) becomes $C_0(x) = 2\Xi'(x)^2$. Thus Theorem 5.1 also implies simplicity of any real Ξ-zero in the finite interval, without assuming simplicity.

7. Failure for a generic positive kernel

The proof uses positivity and decay in its analytic steps, but the numerical moment profile is specific to the Riemann kernel. Consider

$$\Phi_\varepsilon(u) = \psi_\varepsilon(u-1) + \psi_\varepsilon(u+1) + \frac{13}{6}\psi_\varepsilon(u), \qquad \psi_\varepsilon(u) = e^{-2\cosh(u/\varepsilon)}. \tag{31}$$

This kernel is smooth, positive, even, and double-exponentially decreasing. Its Fourier transform factors as

$$\widehat{\Phi_\varepsilon}(z) = 2\varepsilon K_{i\varepsilon z}(2)\left(2\cos z + \frac{13}{6}\right). \tag{32}$$

With $\lambda = \log(3/2)$, one has $2\cosh\lambda = 13/6$, so

$$z = (2k+1)\pi \pm i\lambda$$

are exact nonreal zeros and $0 < \lambda < 1/2$.

The failure appears already in the normalized moments. Let $a = 13/6$ and let $m_{2j}^{(\varepsilon)}(y)$ denote the moments formed from (31). Scaling each of the three bumps and applying dominated convergence gives, for $j \geq 1$,

$$\frac{A_{2j}^{(\varepsilon)}(y)}{\varepsilon^2 M^2} \longrightarrow 4a \operatorname{sinhc} y, \qquad M = \int_{\mathbb{R}} e^{-2\cosh t}\,\mathrm{d}t,$$

while

$$\frac{A_0^{(\varepsilon)}(y)}{\varepsilon^2 M^2} \longrightarrow 8\operatorname{sinhc}(2y) + 4a\operatorname{sinhc} y.$$

Therefore

$$\lim_{\varepsilon\to 0} m_{2j}^{(\varepsilon)}(y) = \frac{a}{2\cosh y + a}. \tag{33}$$

At $y = \lambda$, this limit is $1/2$, which violates the Riemann-kernel bound $m_2(y) < B_2 = 0.0836096840\ldots$ for all small enough ε. At $y = 0$, the limit is $13/25$.

For $0 < \varepsilon \leq 1/4$, the same family also satisfies

$$\Phi_\varepsilon(1/2)^2 < \Phi_\varepsilon(0)\Phi_\varepsilon(1),$$

so it is not log-concave. Neither positivity nor rapid decay supplies the endpoint ratios in (19).

8. The all-orders boundary

The degree-ten theorem does not control the large-frequency sign of C_y. The exact Taylor hierarchy makes the obstruction precise. For $m \geq 0$ set

$$S_m(y,x) = \sum_{j=0}^{2m+1} (-1)^j \frac{m_{2j}(y)x^{2j}}{(2j)!}. \tag{34}$$

The global Taylor inequality $T_{4m+2}(r) < \cos r$ for $r \neq 0$ gives

$$S_m(y,x) < \frac{C_y(x)}{A_0(y)} \qquad (x \neq 0). \tag{35}$$

Proposition 8.1 (Exact hierarchy)**.** *Fix $\varepsilon > 0$ and $R < \infty$. Then S_m converges uniformly to $C_y(x)/A_0(y)$ on*

$$[\varepsilon, 1/2] \times [-R, R].$$

Moreover, $C_y(x) > 0$ on this rectangle if and only if some S_m is uniformly positive there.

Proof. Estimate (5) gives a uniform exponential moment bound

$$\sup_{\varepsilon \le y \le 1/2} \int e^{R|q|} \frac{W_y(u,v)}{A_0(y)}\, \mathrm{d}u\, \mathrm{d}v < \infty.$$

Dominated convergence for the cosine series gives uniform convergence. If C_y/A_0 is positive on the compact rectangle, it has a positive minimum; uniform convergence makes every sufficiently high section positive. The reverse implication follows from (35), with the value at $x = 0$ equal to 1. □

Thus divergence of the positivity radii for the exact sections is equivalent to global open-strip positivity and supplies no weaker intermediate theorem. The cross-endpoint hierarchy is more restrictive. For $\varepsilon \ge 0$, define

$$L_{2j}^{(\varepsilon)} = \frac{A_{2j}(\varepsilon)}{A_0(1/2)}, \qquad U_{2j}^{(\varepsilon)} = \frac{A_{2j}(1/2)}{A_0(\varepsilon)},$$

and

$$\widehat{P}_m^{(\varepsilon)}(x) = 1 + \sum_{\substack{1 \le j \le 2m+1 \\ j \text{ even}}} L_{2j}^{(\varepsilon)} \frac{x^{2j}}{(2j)!} - \sum_{\substack{1 \le j \le 2m+1 \\ j \text{ odd}}} U_{2j}^{(\varepsilon)} \frac{x^{2j}}{(2j)!}. \tag{36}$$

Then

$$\widehat{P}_m^{(\varepsilon)}(x) < S_m(y,x) < \frac{C_y(x)}{A_0(y)}$$

for $x \neq 0$ and $\varepsilon \le y \le 1/2$. Theorem 5.1 is the case $m = 2$, $\varepsilon = 0$.

Let $\widehat{R}_m^{(\varepsilon)}$ be the initial positivity radius of (36). The missing all-orders assertion is

$$\sup_m \widehat{R}_m^{(\varepsilon)} = \infty \qquad \text{for every } \varepsilon > 0. \tag{37}$$

Equation (37) would imply the Riemann Hypothesis. The reverse implication is not established: the lower polynomial combines pessimistic numerator and denominator endpoints attained at different y-values. Therefore (37) may be stronger than RH. The present endpoint identities do not provide the all-orders asymptotics needed to prove it.

9. Reproducibility and certificate scope

The public certificate evaluates Taylor balls for (22) at 0 and 1/2, forms (14), and extracts derivatives through order 12. It uses `python-flint` 0.9.0, FLINT 3.6.0, 269-bit Arb precision, and a formal-series cap of 18. The cap exceeds the thirteen coefficients required through degree 12.

The finite proof obligations are:

1. the twelve endpoint moment balls;
2. the five cross-endpoint ratios;
3. the five Bernstein signs in (24);
4. the five shifted-tail signs in (25);
5. the two rational root-bracket signs.

Arb implements midpoint-radius ball arithmetic with outward enclosures, so a strict comparison succeeds only when the entire output ball lies on the claimed side. The computation uses no quadrature and no list of zeta zeros. The analytic steps linking its finite values to C_y are Lemmas 2.1–4.1 and Proposition 4.2.

The repository records the exact dependency version, output, and SHA-256 checksums. The command is

```
uv run --with python-flint==0.9.0 \
  python code/theta_moment_certificate.py
```

Python optimization must remain disabled because program assertions encode the sign obligations.

10. Conclusion

Endpoint derivatives of ξ at 0 and $1/2$, strict monotonicity of raw moments, and a global cosine minorant produce an explicit lower polynomial for C_y/A_0. Arb certification of its coefficients and root proves the closed finite band in Theorem 5.1. The argument supplies a reproducible local validation of the endpoint-moment method. It does not control the all-orders radii in (37), which remain the first unresolved step toward unbounded open-strip positivity.

Disclosure and licensing

OpenAI Codex assisted with document preparation, computational orchestration, and consistency checks. The named author retains responsibility for the mathematical claims and the released artifacts.

References

[1] G. Csordas, *Fourier transforms of positive definite kernels and the Riemann ξ-function*, Comput. Methods Funct. Theory (2015), doi:10.1007/s40315-014-0105-8.

[2] M. W. Coffey and G. Csordas, *On the log-concavity of a Jacobi theta function*, Math. Comp. **82** (2013), 2265–2272, doi:10.1090/S0025-5718-2013-02681-6.

[3] G. Csordas, T. S. Norfolk, and R. S. Varga, *The Riemann hypothesis and the Turán inequalities*, Trans. Amer. Math. Soc. **296** (1986), 521–541, doi:10.1090/S0002-9947-1986-0846596-4.

[4] G. Csordas and R. S. Varga, *Moment inequalities and the Riemann hypothesis*, Constr. Approx. **4** (1988), 175–198, doi:10.1007/BF02075457.

[5] G. Csordas and R. S. Varga, *Fourier transforms and the Hermite–Biehler theorem*, Proc. Amer. Math. Soc. **107** (1989), 645–652, doi:10.1090/S0002-9939-1989-0982401-1.

[6] G. Csordas and R. S. Varga, *Necessary and sufficient conditions and the Riemann hypothesis*, Adv. in Appl. Math. **11** (1990), 328–357, doi:10.1016/0196-8858(90)90013-O.

[7] D. J. Platt and T. S. Trudgian, *The Riemann hypothesis is true up to* $3 \cdot 10^{12}$, Bull. Lond. Math. Soc. **53** (2021), 792–797, doi:10.1112/blms.12460.